QUASI-FINITENESS AT A PRIME IS PRESERVED BY A SURJECTIVE CHANGE OF THE FOUR RINGS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement contains an Xy-pic diagram; the preamble spans of this card run 8-18 of the chapter's own preamble file and therefore include the Xy-pic package and option setup at lines 15-18.

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Allowed standard prerequisites (editorial).

- Isolated points of a fibre give a product decomposition. Invoked at the start of the counted proof (lines 30527-30531). Writes the fibre as a product with a finite factor containing the point.
- Finiteness and dimension zero detected after a product decomposition. Used at the end of the counted proof (lines 30545-30554). Shows the point stays isolated in the new fibre.

Source reference. Original lemma 00PJ: sources/stacks-algebra/pinned-original-source.tex, source lines 30323–30323. Navigation pointer for \ref{lemma-isolated-point} used inside the counted span (line 30530); the argument that consumes it is retained in the span.

Source reference. Original lemma 00ED: sources/stacks-algebra/pinned-original-source.tex, source lines 4039–4039. Navigation pointer for \ref{lemma-spec-product} used inside the counted span (line 30550); the argument that consumes it is retained in the span.

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Source reference. Original lemma 00KJ: sources/stacks-algebra/pinned-original-source.tex, source lines 14632–14632. Navigation pointer for \ref{proposition-dimension-zero-ring} used inside the counted span (line 30553); the argument that consumes it is retained in the span.

Lemma 0.1. *Let*

$$\begin{array}{ccc} S & \longrightarrow & S' \\ \uparrow & & \uparrow \\ R & \longrightarrow & R' \end{array} \qquad \begin{array}{ccc} \mathfrak{q} & \longrightarrow & \mathfrak{q}' \\ \downarrow & & \downarrow \\ \mathfrak{p} & \longrightarrow & \mathfrak{p}' \end{array}$$

be a commutative diagram of rings with primes as indicated. Assume $R \to S$ of finite type, and $S \otimes_R R' \to S'$ surjective. If $R \to S$ is quasi-finite at $\mathfrak{q}$, then $R' \to S'$ is quasi-finite at $\mathfrak{q}'$.

Proof. Write $S \otimes_R \kappa(\mathfrak{p}) = S_1 \times S_2$ with S_1 finite over $\kappa(\mathfrak{p})$ and such that $\mathfrak{q}$ corresponds to a point of S_1 as in Lemma 00PJ. This product decomposition induces a corresponding product decomposition for any $S \otimes_R \kappa(\mathfrak{p})$-algebra. In particular, we obtain $S' \otimes_{R'} \kappa(\mathfrak{p}') = S'_1 \times S'_2$. Because $S \otimes_R R' \to S'$ is surjective the canonical map $(S \otimes_R \kappa(\mathfrak{p})) \otimes_{\kappa(\mathfrak{p})} \kappa(\mathfrak{p}') \to S' \otimes_{R'} \kappa(\mathfrak{p}')$ is surjective and hence $S_i \otimes_{\kappa(\mathfrak{p})} \kappa(\mathfrak{p}') \to S'_i$ is surjective. It follows that S'_1 is finite over $\kappa(\mathfrak{p}')$. The map $S' \otimes_{R'} \kappa(\mathfrak{p}') \to \kappa(\mathfrak{q}')$ factors through S'_1 (i.e. it annihilates the factor S'_2) because the map $S \otimes_R \kappa(\mathfrak{p}) \to \kappa(\mathfrak{q})$ factors through S_1 (i.e. it annihilates the factor S_2). Thus $\mathfrak{q}'$ corresponds to a point of $\mathrm{Spec}(S'_1)$ in the disjoint union decomposition of the fibre: $\mathrm{Spec}(S' \otimes_{R'} \kappa(\mathfrak{p}')) = \mathrm{Spec}(S'_1) \amalg \mathrm{Spec}(S'_2)$, see Lemma 00ED. Since S'_1 is finite over a field, it is Artinian ring, and hence $\mathrm{Spec}(S'_1)$ is a finite discrete set. (See Proposition 00KJ.) We conclude $\mathfrak{q}'$ is isolated in its fibre as desired. □